The radioactive elements were called *radioelements*. Lacking names for these radioelements, letters such as X, Y, Z, A, B, etc., were added to the symbol for the primary (i.e. parent) element. Thus, UX was produced from the radioactive decay of uranium, ThX from that of thorium, etc. These new radioelements (UX, ThX, etc.) had chemical properties that were different from the original elements, and could be separated from them through chemical processes such as precipitation, volatilization, electrolytic deposition, etc. The radioactive daughter elements decayed further to form still other elements, symbolized as UY, ThA, etc. A typical decay chain could be written: Ra → Rn → RaA → RaB → , etc.; see Fig. 1.2.

A careful study of the radiation emitted from these radioactive elements demonstrated that it consisted of three components which were given the designation alpha (α), beta (β), and gamma (γ). *Alpha-radiation* was shown to be identical to helium ions, whereas *beta-radiation* was identical to electrons. *Gamma-radiation* had the same electromagnetic nature as X-rays but was of higher energy. The rate of radioactive decay per unit weight was found to be fixed for any specific radioelement, no matter what its chemical or physical state was, though this rate differed greatly for different radioelements. The decay rate could be expressed in terms of a *half-life*, which is the time it takes for the radioactivity of a radioelement to decay to one-half of its original value. Half-lives for the different radioelements were found to vary from fractions of a second to millions of years; e.g. that of ThA is 0.1 of a second, of UX it is 24.1 days (Fig. 1.1), and of uranium, millions of years.

1.3. DISCOVERY OF ISOTOPES

By 1910 approximately 40 different chemical species had been identified through their chemical nature, the properties of their radiation, and their characteristic half-lives. The study of the generic relationships in the decay of the radioactive species showed that the radioelements could be divided into three distinct series. Two of these originated in uranium and the third in thorium. B. Boltwood found that all three of the series ended in the same element — lead.

A major difficulty obvious to scientists at that time involved the fact that while it was known from the Periodic Table (Appendix I) that there was space for only 11 elements between lead and uranium, approximately 40 radioelements were known in the decay series from uranium to lead. To add to the confusion was the fact that it was found that in many cases it was not possible to separate some of the radioelements from each other by normal chemical means. For example, the radioelement RaD was found to be chemically identical to lead. In a similar manner, spectrographic investigations of the radioelement ionium showed exactly the same spectral lines that had been found previously to be due to the element thorium.

In 1913 K. Fajans and Soddy independently provided the explanation for these seemingly contradictory conditions. They stated that by the radioactive α-decay a new

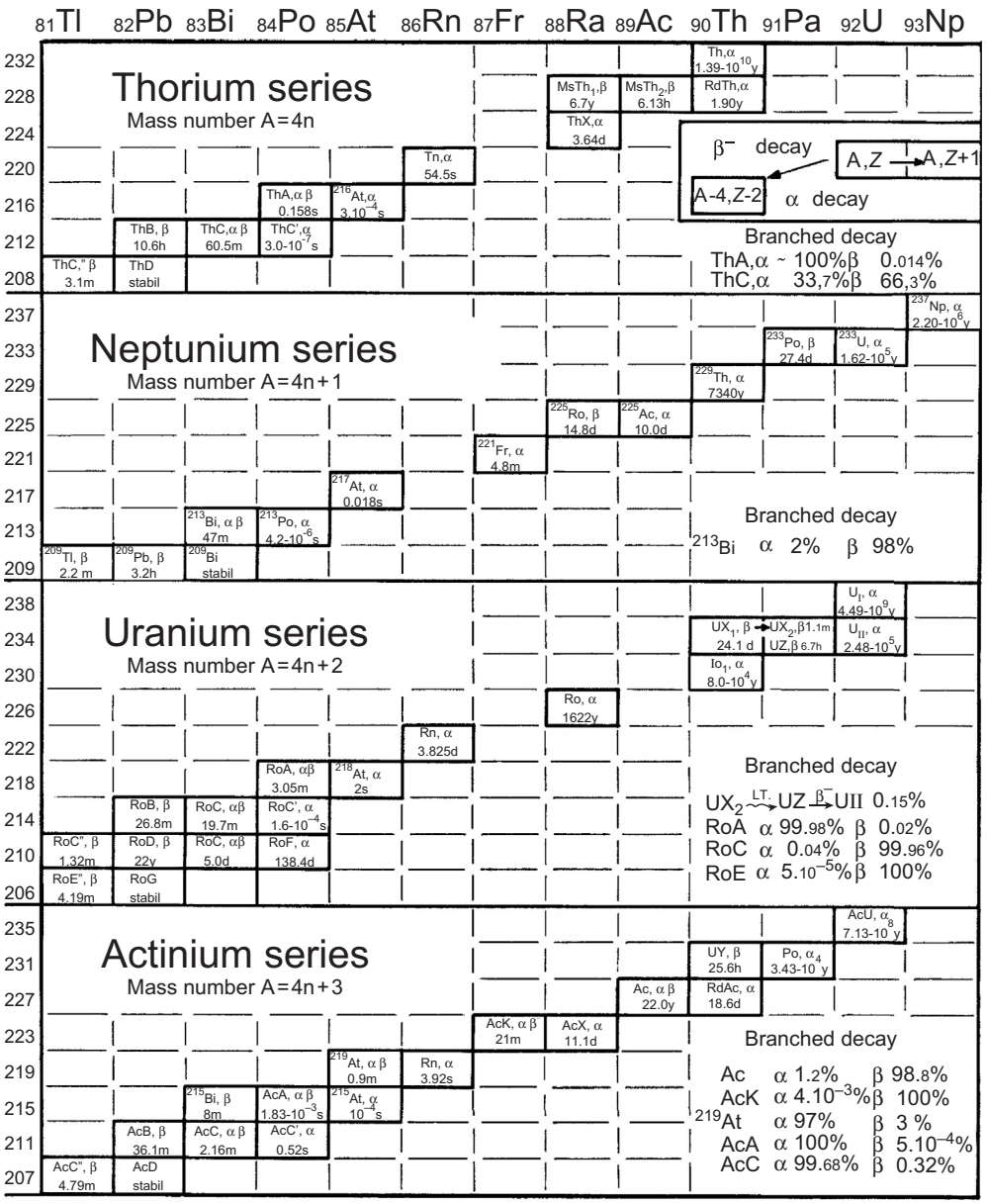


Figure 1.2 The three naturally occurring radioactive decay series and the man-made neptunium series. Although ^{239}Pu (which is the parent to the actinium series) and ^{244}Pu (which is the parent to the thorium series) have been discovered in nature, the decay series shown here begin with the most abundant long-lived nuclides.

element is produced two places to the left of the mother element in the periodic system and in α-decay a new element is produced one place to the right of the mother element (Fig. 1.2). The radioelements that fall in the same place in the periodic system are chemically identical. Soddy proposed the name *isotopes* to account for different radioactive species which have the same chemical identity.

Research by J. J. Thomson soon provided conclusive support for the existence of isotopes. If a beam of positively charged gaseous ions is allowed to pass through electric or magnetic fields, the ions follow hyperbolic paths which are dependent on the masses and charges of the gaseous ions (see Fig. 3.1 and associated text). When these ion beams strike photographic plates, a darkening results which is proportional to the number of ions which hit the plate. By using this technique with neon gas, Thomson found that neon consists of two types of atoms with different atomic masses. The mass numbers for these two isotopes were 20 and 22. Moreover, from the degree of darkening of the photographic plate, Thomson calculated that neon consisted to about 90% of atoms with mass number 20, and 10% of atoms with mass number 22.

Thus a chemical element may consist of several kinds of atoms with different masses but with the same chemical properties. The 40 radioelements were, in truth, not 40 different elements but were isotopes of the 11 different chemical elements from lead to uranium.

To specify a particular isotope of an element, the *atomic number* (i.e. order, number, or place in the Periodic Table of elements) is written as a subscript to the left of the chemical symbol and the mass number (i.e. the integer value nearest to the mass of the neutral atom, measured in atomic weight units) as a superscript to the left. Thus the isotope of uranium with mass number 238 is written as ${}^{238}_{92}\mathrm{U}$. Similarly, the isotope of protactinium with mass number 234 is designated ${}^{234}_{91}\mathrm{Pa}$. For an alpha-particle we use either the Greek letter α or ${}^{4}_{2}\mathrm{He}$. Similarly, the beta-particle is designated either by the Greek letter β or by the symbol ${}^{0}_{-1}\mathrm{e}$, and the electron antineutrino is designated as $\overline{\nu}$. However, the neutrino emitted is usually not written out in β-decay reactions like (1.2).

In radioactive decay both mass number and atomic number are conserved. Thus in the decay chain of the first two steps are written:

$$ {}^{238}_{92}\mathrm{U} \;\rightarrow\; {}^{234}_{90}\mathrm{Th} + {}^{4}_{2}\mathrm{He} \tag{1.1} $$

$$ {}^{234}_{90}\mathrm{Th} \;\rightarrow\; {}^{234}_{91}\mathrm{Pa} + {}^{0}_{-1}\mathrm{e} + \overline{\nu} \tag{1.2} $$

Frequently, in such a chain, the half-life ($t_{1/2}$) for the radioactive decay is shown either above or below the arrow. A shorter notation is commonly used:

$$ {}^{238}\mathrm{U} \xrightarrow[4.5\times10^{9}\mathrm{y}]{\alpha} {}^{234}\mathrm{Th} \xrightarrow[24\ \mathrm{d}]{\beta^-} {}^{234}\mathrm{Pa} \xrightarrow[1.1\ \mathrm{min}]{\beta^-} {}^{234}\mathrm{U} \xrightarrow[2.5\times10^{5}\mathrm{y}]{\alpha} {}^{230}\mathrm{Th},\ \text{etc.} \tag{1.3} $$

where the half-lives are given in years (y) and days (d)[2]. The three naturally occurring radioactive decay series, which are known as the *thorium series*, the *uranium series*, and the *actinium series*, are shown in Fig. 1.2. A fourth series, which originates in the synthetic element neptunium, is also shown. This series is not found naturally on earth since all of the radioactive species have decayed away long ago. Both the present symbolism of the isotope as well as the historical (i.e. "radioelement") symbolism are given in Fig. 1.2. Note that the *rule of Fajans and Soddy* is followed in each series so that α-decay causes a decrease in atomic number by two units and mass number by four, whereas β-decay produces no change in mass number but an increase in atomic number by one unit. Moreover, we see a pattern occurring frequently in these series where an α-decay step is followed by two β-decay steps. All known isotopes of elements $_{81}$Tl to $_{92}$U are given in Figure 13.4.

1.4. ATOMIC MODELS

Neither radioactive decay nor the discovery of isotopes provided information on the internal structure of atoms. Such information was obtained from scattering experiments in which a substance, such as a thin metal foil, was irradiated with a beam of ⟨-particles and the intensity (measured by counting scintillations from the scattered particles hitting a fluorescent screen) of the particles scattered at different angles measured (see Fig. 10.4). It was assumed that the deflection of the particles was caused by collisions with the atoms of the irradiated material. About one in 8000 of the á-particles was strongly deflected through angles greater than 90°. Consideration of these rare events led Rutherford in 1911 to the conclusion that the entire positive charge of an atom must be concentrated in a very small volume whose diameter is about 10^{-14} m. This small part of the atom he called the *nucleus*. The atomic electrons have much smaller mass and were assumed to surround the nucleus. The total atom with the external electrons had a radius of approximately 10^{-10} m in contrast to the much smaller radius calculated for the nucleus.

It was soon shown that the positive charge of the atomic nucleus was identical to the atomic number assigned to an element in the periodic system of Mendeleev. The conclusion, then, is that in a neutral atom the small, positively charged nucleus was surrounded by electrons whose number was equal to the total positive charge of the nucleus. In 1913 N. Bohr, using quantum mechanical concepts, proposed such a model of the atom which remains the basis of the modern atomic theory.

Our understanding of the nucleus has grown rapidly since Rutherford's scattering experiments. Some of the important steps in the history of nuclear science are listed in

[2] IUPAC recommends a for *annum*, instead of y, however y will be used throughout this text as it remains the commonly used term.